Experiment 1 ? Program 2

Data Science - 22DS102006

2. Interacting with Web APIs and Databases

a. GitHub Issues to DataFrame

```
import requests
import pandas as pd

url = 'https://api.github.com/repos/pandas-dev/pandas/issues'
response = requests.get(url)
issues = response.json()

df_issues = pd.DataFrame(issues)
print(df_issues[['id', 'title', 'state', 'created_at']].head())
```

/*

Step by Step Explanation

requests.get(url): Sends a GET request to GitHub's API to retrieve the issues of the pandas-dev/pandas repository.

.json(): Parses the JSON response into a Python object (usually a list of dictionaries).

pd.DataFrame(issues): Converts the list of issue dictionaries into a DataFrame.

print(...): Displays the first 5 issues, showing selected columns: id, title, state, and created_at.

*/

b. SQLite Database Operations

```
import sqlite3
import pandas as pd

# Connect to database
conn = sqlite3.connect("students.db")
cursor = conn.cursor()

# Create table
cursor.execute("'CREATE TABLE IF NOT EXISTS students (id INTEGER, name TEXT)"')

# Insert data
```

```
cursor.execute("INSERT INTO students VALUES (1, 'John')")
conn.commit()

cursor.execute("INSERT INTO students VALUES (2, 'Joy')")
conn.commit()

# Read data
df = pd.read_sql_query("SELECT * FROM students", conn)
print(df)

# Update and Delete
cursor.execute("UPDATE students SET name = 'Alice' WHERE id = 1")
cursor.execute("DELETE FROM students WHERE id = 1")
conn.commit()
conn.close()

/*
```

Explanation

sqlite3.connect(...): Connects to a database file (students.db). If it doesn’t exist, it creates one.

cursor.execute(...): Executes SQL queries.

CREATE TABLE IF NOT EXISTS: Ensures the table is created only once.

INSERT INTO: Adds a new student with id = 1, name = John.

pd.read_sql_query(...): Reads the table as a DataFrame for easy display and analysis.

UPDATE: Changes the student's name to Alice.

DELETE: Removes the student record from the table.

conn.commit(): Saves changes to the database.

conn.close(): Closes the connection.

*/